

Project Design Phase Solution Architecture

Date	15 February 2025
Team ID	
Project Name	
Maximum Marks	4 Marks

Solution Architecture

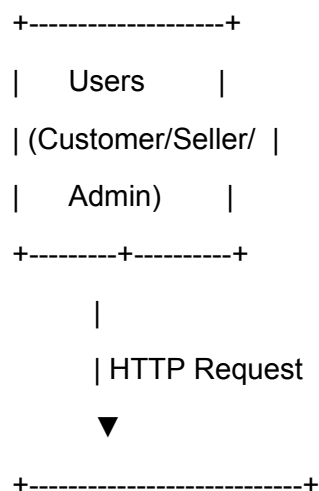
The **Book Store Web Application** is developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**. The system provides an online platform where users can browse, search, and purchase books, while sellers manage books and administrators control the overall system.

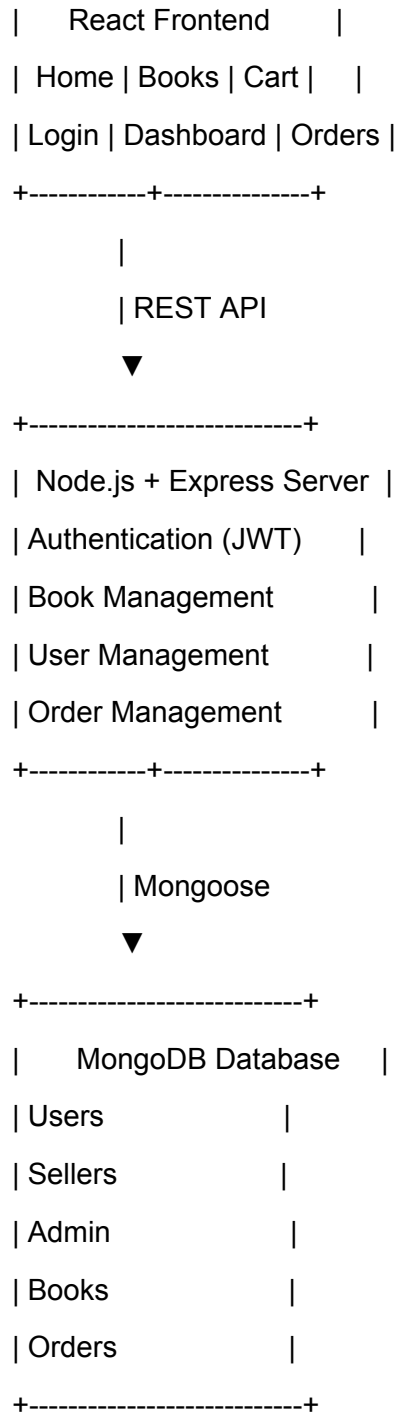
Goals of the Solution

- Provide an easy and secure online book shopping platform.
 - Enable users to search, view, and purchase books conveniently.
 - Help sellers manage book inventory and customer orders.
 - Allow administrators to monitor users, sellers, books, and orders.
 - Ensure a responsive, scalable, and user-friendly application.
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Solution Architecture Diagram

BOOK STORE WEB APPLICATION





Architecture Flow

1. Users access the Book Store website using a web browser.
2. The React frontend sends requests to the Express backend.
3. The backend authenticates users using JWT.
4. The backend processes requests such as login, book search, and order placement.

5. MongoDB stores and retrieves user, book, seller, and order information.
6. The processed data is sent back to the frontend and displayed to the user.

Technology Stack

- **Frontend:** React.js
- **Backend:** Node.js & Express.js
- **Database:** MongoDB
- **Authentication:** JWT
- **Version Control:** Git & GitHub

